

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: **Data Analysis with SAS / SPSS / R**

PRACTICAL 4

AIM:Applying conditional filters subset() or filter() in R.

>COMMANDS

```
install.packages("dplyr")
```

```
library(dplyr)
```

```
library(readr)
```

```
# CORRECTION: Reading the correct file
```

```
job_title <-
```

```
read_csv("AI_Impact_on_Jobs_2030.csv")
```

```
head(job_title)
```

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```
# Base R subset() examples
# Example 1: High Salary (> 100,000)
high_salary_subset <- subset(job_title,
Average_Salary > 100000)
cat("Number of high-salary jobs (> 100,000):",
nrow(high_salary_subset), "\n")
summary(high_salary_subset$Average_Salary)

# Example 2: High Salary (> 100,000) AND
Senior Experience (> 20 years)
low_crime_high_value_subset <-
subset(job_title, Average_Salary > 100000 &
Years_Experience > 20)
```

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```
cat("Number of high-salary, senior jobs:",  
nrow(low_crime_high_value_subset), "\n")  
head(low_crime_high_value_subset)
```

```
# Example 3: High Risk OR Low Experience (<  
5 years)
```

```
special_jobs_subset <- subset(job_title,  
Risk_Category == "High" | Years_Experience <  
5)
```

```
cat("Number of high-risk or low-experience  
jobs:", nrow(special_jobs_subset), "\n")  
head(special_jobs_subset)
```

```
# dplyr filter() examples
```

```
# Example 1: Low Automation Risk (< 0.2)
```

```
low_automation_filter <- job_title |>
```

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```
filter(Automation_Probability_2030 < 0.2)
```

```
cat("Number of jobs with low automation risk  
(< 0.2):", nrow(low_automation_filter), "\n")  
summary(low_automation_filter$Automation_  
Probability_2030)
```

```
# Example 2: Not Low Risk AND High AI  
Exposure (> 0.8)  
high_aiexp_not_lowrisk_filter <- job_title |>  
  filter(Risk_Category != "Low",  
  AI_Exposure_Index > 0.8)
```

```
cat("Number of non-Low Risk, high AI  
exposure jobs:",  
nrow(high_aiexp_not_lowrisk_filter), "\n")
```

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```
head(high_aiexp_not_lowrisk_filter)

# Example 3: Specific Experience Years (10,
15, or 20 years)
rad_10_15_20_filter <- job_title |>
  filter(Years_Experience %in% c(10, 15, 20))

cat("Number of jobs with 10, 15, or 20 years
experience:", nrow(rad_10_15_20_filter), "\n")
table(rad_10_15_20_filter$Years_Experience)
```

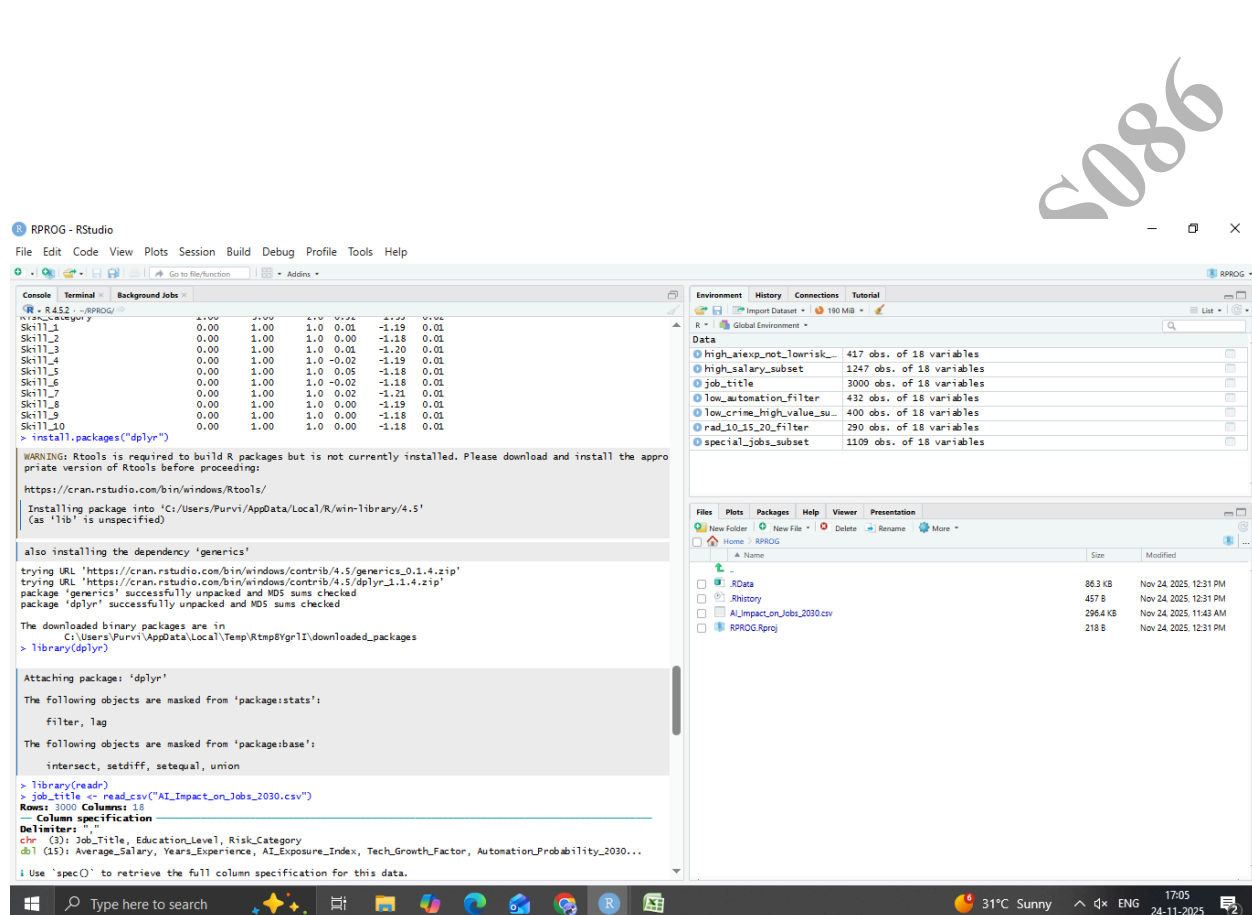
OUTPUT:

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The screenshot displays the RStudio environment. The console shows the installation of the 'dplyr' package and the loading of a dataset named 'job_title'. The environment pane on the right lists several data objects, including 'HighSalary_notLowRisk...', 'HighSalary_subset', 'job_title', 'LowAutomation_Filter', 'LowCrime_HighValue_Su...', 'rad_10_15_20_Filter', and 'special_jobs_subset'. The console output includes a warning about Rtools, the installation of 'dplyr', and the successful loading of the 'job_title' dataset with 3000 rows and 18 columns.

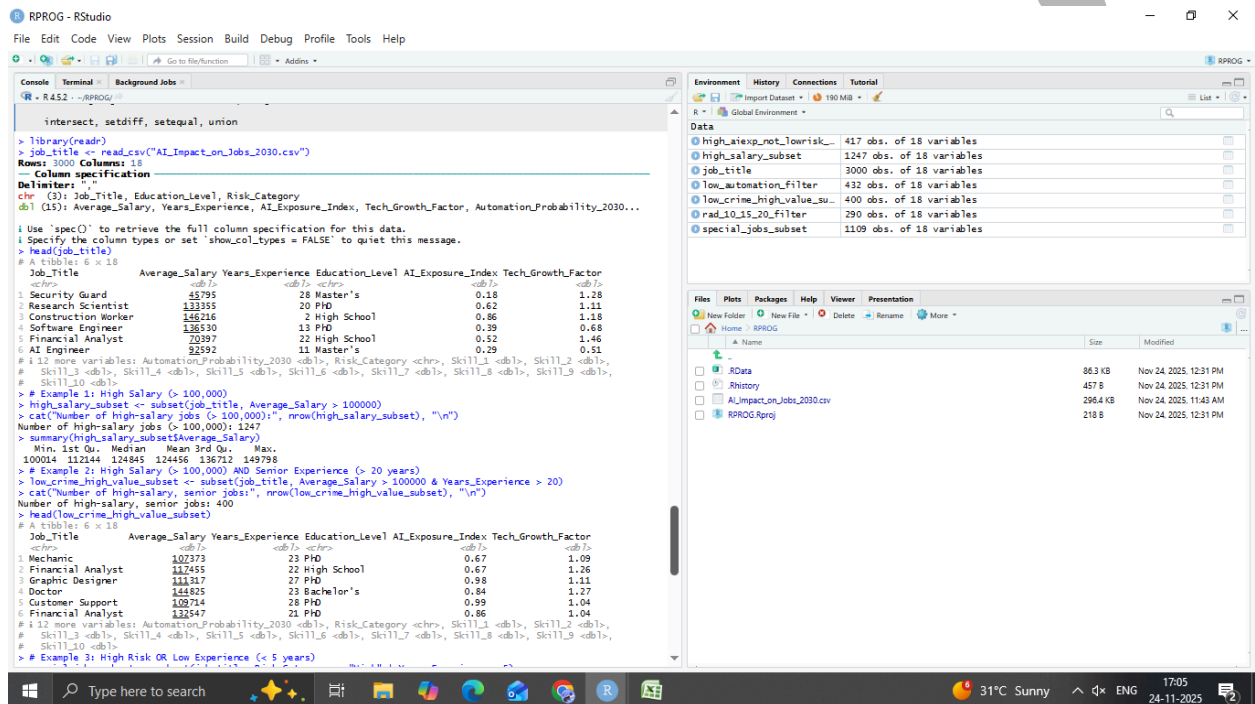
```
R - R4.2.2 -- RPROG --  
File Edit Code View Plots Session Build Debug Profile Tools Help  
Console Terminal Background Jobs  
R - R4.2.2 -- RPROG --  
install.packages("dplyr")  
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:  
https://cran.rstudio.com/bin/windows/Rtools/  
Installing package into 'C:/Users/Purvi/AppData/Local/R/win-library/4.5'  
(as 'lib' is unspecified)  
also installing the dependency 'generics'  
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/generics_0.1.4.zip'  
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/dplyr_1.1.4.zip'  
package 'generics' successfully unpacked and MD5 sums checked  
package 'dplyr' successfully unpacked and MD5 sums checked  
The downloaded binary packages are in  
C:/Users/Purvi/AppData/Local/Temp/Rtmp8Ygr1Z/downloaded_packages  
> library(dplyr)  
Attaching package: 'dplyr'  
The following objects are masked from 'package:stats':  
filter, lag  
The following objects are masked from 'package:base':  
intersect, setdiff, setequal, union  
> library(readr)  
> job_title <- read_csv("AI_Impact_on_Jobs_2030.csv")  
Rows: 3000 Columns: 18  
-- Column specification --  
Delimiter: ","  
chr (3): Job_Title, Education_Level, Risk_Category  
dbl (15): Average_Salary, Years_Experience, AI_Exposure_Index, Tech_Growth_Factor, Automation_Probability_2030...  
i Use 'spec()' to retrieve the full column specification for this data.  
Environment History Connections Tutorial  
R - Global Environment  
Data  
HighSalary_notLowRisk... 417 obs. of 18 variables  
HighSalary_subset 1247 obs. of 18 variables  
job_title 3000 obs. of 18 variables  
LowAutomation_Filter 432 obs. of 18 variables  
LowCrime_HighValue_Su... 400 obs. of 18 variables  
rad_10_15_20_Filter 290 obs. of 18 variables  
special_jobs_subset 1109 obs. of 18 variables  
Files Plots Packages Help Viewer Presentation  
New Folder New File Delete Rename More  
Home RPROG  
Name Size Modified  
RData 863 KB Nov 24, 2025, 12:31 PM  
Rhistory 457 B Nov 24, 2025, 12:31 PM  
AI_Impact_on_Jobs_2030.csv 296.4 KB Nov 24, 2025, 11:43 AM  
RPROG.Rproj 218 B Nov 24, 2025, 12:31 PM  
Type here to search 31°C Sunny 17:05 24-11-2025
```

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The screenshot displays the RStudio environment. The console shows the following R code and output:

```
intersect, setdiff, setequal, union

> library(readr)
> job_title <- read_csv("AI_Impact_on_Jobs_2030.csv")
Rows: 3000 Columns: 18
Column specification:
Delimiter:
chr (3): Job_Title, Education_Level, Risk_Category
dbl (15): Average_Salary, Years_Experience, AI_Exposure_Index, Tech_Growth_Factor, Automation_Probability_2030...

i Use 'spec()' to retrieve the full column specification for this data.
i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
> head(job_title)
# A tibble: 6 x 18
  Job_Title      Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
  <chr>          <dbl>          <dbl>          <chr>          <dbl>          <dbl>
1 Security Guard  45795          28 Master's      0.18          1.28
2 Research Scientist 133355         20 PhD          0.62          1.11
3 Construction Worker 146216         2 High School    0.86          1.18
4 Software Engineer  136530         13 PhD          0.39          0.68
5 Financial Analyst  70397         22 High School  0.52          1.46
6 AI Engineer      92592         11 Master's      0.29          0.51

# i 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
# # Example 1: High Salary (> 100,000)
# high_salary_subset <- subset(job_title, Average_Salary > 100000)
> cat("Number of high-salary jobs (> 100,000):", nrow(high_salary_subset), "\n")
Number of high-salary jobs (> 100,000): 1247
> summary(high_salary_subset$Average_Salary)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
100014 112244 124845 124456 136712 149798
# # Example 2: High Salary (> 100,000) AND Senior Experience (> 20 years)
# low_crime_high_value_subset <- subset(job_title, Average_Salary > 100000 & Years_Experience > 20)
> cat("Number of high-salary, senior jobs:", nrow(low_crime_high_value_subset), "\n")
Number of high-salary, senior jobs: 400
> head(low_crime_high_value_subset)
# A tibble: 6 x 18
  Job_Title      Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
  <chr>          <dbl>          <dbl>          <chr>          <dbl>          <dbl>
1 Mechanic      102373         23 PhD          0.67          1.09
2 Financial Analyst 117455         22 High School  0.67          1.36
3 Graphic Designer 111317         27 PhD          0.98          1.11
4 Doctor        144825         23 Bachelor's  0.84          1.27
5 Customer Support 109714         28 PhD          0.99          1.04
6 Financial Analyst 132547         21 PhD          0.86          1.04

# i 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
# # Example 3: High Risk OR Low Experience (< 5 years)
```

The Environment pane on the right shows the following data objects:

Object	Size	Modified
High_Alexp_not_lowrisk...	417 obs. of 18 variables	
high_salary_subset	1247 obs. of 18 variables	
job_title	3000 obs. of 18 variables	
low_automation_filter	432 obs. of 18 variables	
low_crime_high_value_su...	400 obs. of 18 variables	
rad_10_15_20_filter	290 obs. of 18 variables	
special_jobs_subset	1109 obs. of 18 variables	

The Files pane shows the following files:

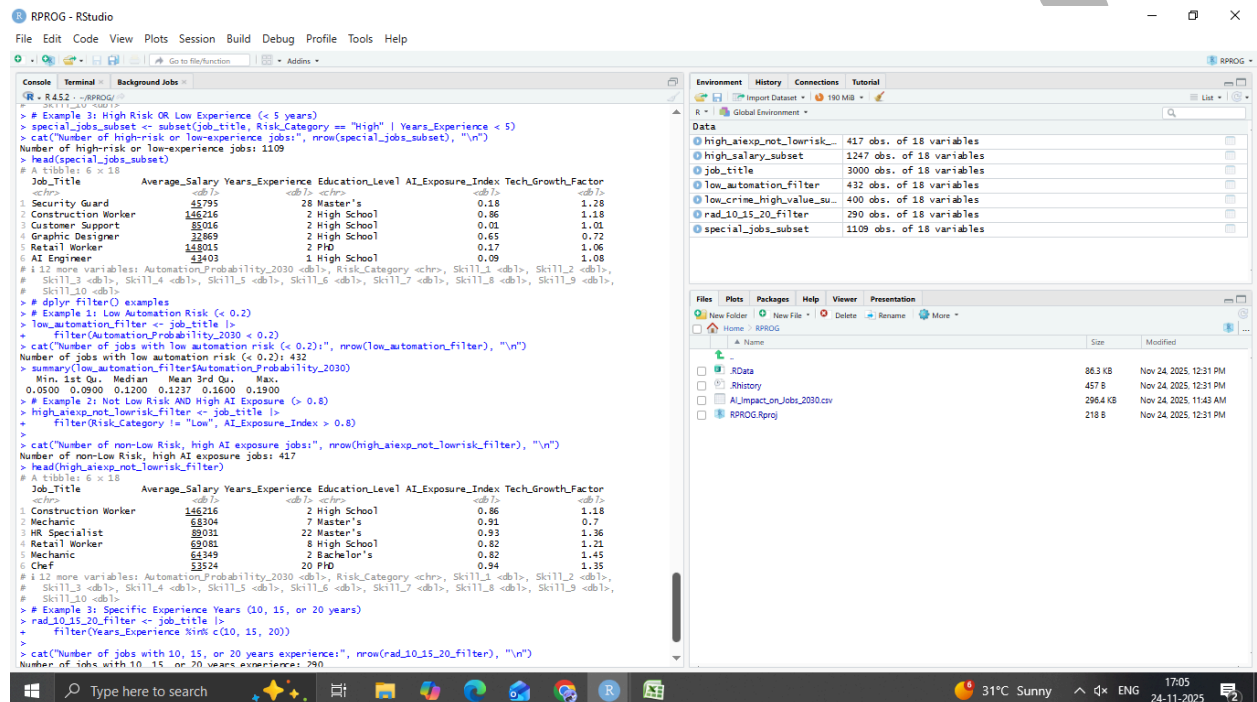
Name	Size	Modified
RData	863 KB	Nov 24, 2025, 12:31 PM
Rhistory	457 B	Nov 24, 2025, 12:31 PM
AI_Impact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
RPROG.Rproj	218 B	Nov 24, 2025, 12:31 PM

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The screenshot displays the RStudio environment with the following components:

- Console:** Shows R code execution for filtering data based on job risk and automation probability. The code includes comments and variable assignments like `special_jobs_subset` and `low_automation_filter`. It also displays summary statistics for the filtered datasets.
- Environment:** Lists the objects created in the R session, including `high_ai_exp_not_low_risk_subset`, `high_salary_subset`, `job_title`, `low_automation_filter`, `low_crime_high_value_subset`, `rad_10_15_20_filter`, and `special_jobs_subset`.
- Files:** Shows the file explorer with a folder named `RPROG` containing files like `RData`, `Rhistory`, `AI_Impact_on_Jobs_2020.csv`, and `RPROG.Rproj`.

The code in the console includes the following snippets:

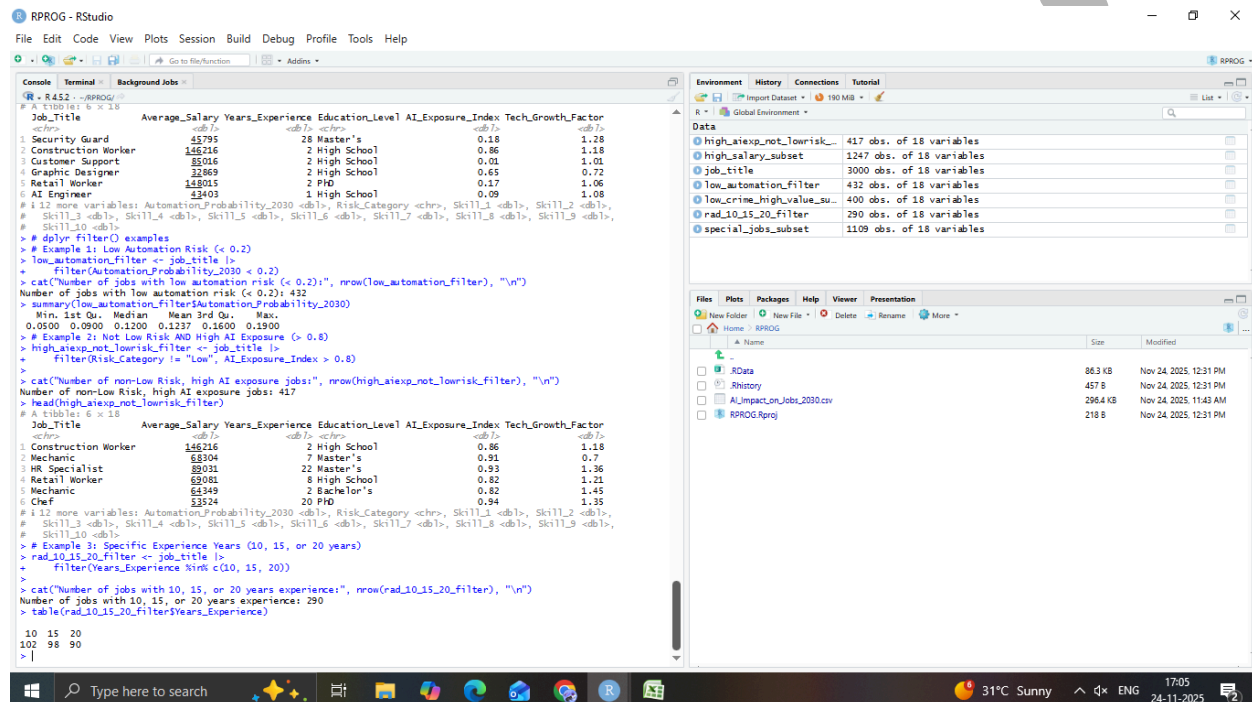
```
> # Example 3: High Risk OR Low Experience (< 5 years)
> special_jobs_subset <- subset(job_title, Risk_Category == "High" | Years_Experience < 5)
> cat("Number of high-risk or low-experience jobs:", nrow(special_jobs_subset), "\n")
Number of high-risk or low-experience jobs: 1109
> head(special_jobs_subset)
# A tibble: 6 x 18
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
  <chr> <dbl> <dbl> <chr> <dbl> <dbl>
1 Security Guard 42795 20 Master's 0.86 1.28
2 Construction Worker 146216 2 High School 0.86 1.18
3 Customer Support 82016 2 High School 0.01 1.01
4 Graphic Designer 32869 2 High School 0.65 0.72
5 Retail Worker 148015 2 PhD 0.17 1.06
6 AI Engineer 12403 1 High School 0.09 1.08
# 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
> # dplyr::filter() examples
> # Example 1: Low Automation Risk (< 0.2)
> low_automation_filter <- job_title |>
+ filter(Automation_Probability_2030 < 0.2)
> cat("Number of jobs with low automation risk (< 0.2):", nrow(low_automation_filter), "\n")
Number of jobs with low automation risk (< 0.2): 432
> summary(low_automation_filtersAutomation_Probability_2030)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0500 0.0900 0.1200 0.1237 0.1600 0.1900
> # Example 2: Not Low Risk AND High AI Exposure (> 0.8)
> high_ai_exp_not_low_risk_filter <- job_title |>
+ filter(Risk_Category != "Low", AI_Exposure_Index > 0.8)
> cat("Number of non-Low Risk, high AI exposure jobs:", nrow(high_ai_exp_not_low_risk_filter), "\n")
Number of non-Low Risk, high AI exposure jobs: 417
> head(high_ai_exp_not_low_risk_filter)
# A tibble: 6 x 18
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
  <chr> <dbl> <dbl> <chr> <dbl> <dbl>
1 Construction Worker 146216 2 High School 0.86 1.18
2 Mechanic 68304 7 Master's 0.91 0.7
3 HR Specialist 82031 22 Master's 0.93 1.36
4 Retail Worker 69081 8 High School 0.82 1.21
5 Mechanic 63449 2 Bachelor's 0.82 1.45
6 Chef 53524 20 PhD 0.94 1.35
# 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
> # Example 3: Specific Experience Years (10, 15, or 20 years)
> rad_10_15_20_filter <- job_title |>
+ filter(Years_Experience %in% c(10, 15, 20))
> cat("Number of jobs with 10, 15, or 20 years experience:", nrow(rad_10_15_20_filter), "\n")
Number of jobs with 10, 15, or 20 years experience: 290
```

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SUBJECT NAME: Data Analysis with SAS / SPSS / R



The screenshot displays the RStudio environment with the following components:

- Console:** Shows R code execution for data analysis, including data loading, filtering, and summarization. The code includes comments in Hindi and English, and the output shows a summary of jobs with low automation risk.
- Environment:** Lists the objects created in the R session, including 'High_Aiexp_not_lowrisk...', 'High_salary_subset', 'job_title', 'Low_automation_filter', 'Low_crime_high_value_su...', 'rad_10_15_20_filter', and 'special_jobs_subset'.
- Files:** Shows the file explorer with a list of files, including 'RData', 'Rhistory', 'AI_Impact_on_Jobs_2030.csv', and 'RPROG.Rproj'.

```
# A tibble: 6 x 18
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
<chr> <dbl> <dbl> <chr> <dbl> <dbl>
1 Security Guard 45795 28 Master's 0.18 1.28
2 Construction Worker 146216 2 High School 0.86 1.18
3 Customer Support 88016 2 High School 0.01 1.01
4 Graphic Designer 32869 2 High School 0.65 0.72
5 Retail Worker 148015 2 PhD 0.17 1.06
6 AI Engineer 42403 1 High School 0.09 1.08
# i 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
# Example 1: Low Automation Risk (< 0.2)
> low_automation_filter <- job_title |>
+ filter(Automation_Probability_2030 < 0.2)
> cat("Number of jobs with Low automation risk (< 0.2):", nrow(low_automation_filter), "\n")
Number of jobs with Low automation risk (< 0.2): 432
> summary(low_automation_filter$Automation_Probability_2030)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0500 0.0900 0.1200 0.1237 0.1600 0.1900
# Example 2: Not Low Risk AND High AI Exposure (> 0.8)
> high_aiexp_not_lowrisk_filter <- job_title |>
+ filter(Risk_Category != "Low", AI_Exposure_Index > 0.8)
> cat("Number of non-Low Risk, high AI exposure jobs:", nrow(high_aiexp_not_lowrisk_filter), "\n")
Number of non-Low Risk, high AI exposure jobs: 417
> head(high_aiexp_not_lowrisk_filter)
# A tibble: 6 x 18
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
<chr> <dbl> <dbl> <chr> <dbl> <dbl>
1 Construction Worker 146216 2 High School 0.86 1.18
2 Mechanic 69304 7 Master's 0.91 0.7
3 HR Specialist 88031 22 Master's 0.93 1.36
4 Retail Worker 69081 8 High School 0.82 1.21
5 Mechanic 69349 2 Bachelor's 0.82 1.45
6 Chef 53524 20 PhD 0.94 1.35
# i 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
# Example 3: Specific Experience Years (10, 15, or 20 years)
> rad_10_15_20_filter <- job_title |>
+ filter(Years_Experience %in% c(10, 15, 20))
> cat("Number of jobs with 10, 15, or 20 years experience:", nrow(rad_10_15_20_filter), "\n")
Number of jobs with 10, 15, or 20 years experience: 290
> table(rad_10_15_20_filter$Years_Experience)
  10 15 20
102 98 90
```

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